SUPPLY CHAIN PERFORMANCE REPORT

End-to-End Supply Chain Analysis

Dataset	DataCo Supply Chain Dataset
Orders Analyzed	172,765
Analysis Scope	Delivery, profitability, bottlenecks & ML
Model	Random Forest with SMOTE
Academic Year	2026-2027
Subject	Data Visualization
Performed by	Vishal Tilokani
Problem Statement	Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to buyers at point of purchase.

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1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers 172,765 orders, focusing on delivery performance, profitability, operational bottlenecks, temporal patterns, and predictive risk identification.

The headline finding is stark: 54.71% of all orders are delivered late, while delayed orders are associated with approximately $2.1M in profit at risk. Total profit from profitable orders reaches $7.5M. A Random Forest predictive model achieved 74% accuracy in identifying at-risk orders before they become late.

Key Finding	Metric
Overall Late Delivery Rate	**54.71%**
On-Time Delivery Rate	**45.29%**
Total Orders Analyzed	**172,765**
Total Profit (profitable orders)	**$7.5M**
Profit at Risk (delayed orders)	**$2.1M**
Predictive Model Accuracy (RF)	**74%**

2. Business Context & Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to buyers at point of purchase.

Analytical Objectives

- Understand delivery performance across region, mode, time and customer segment.
- Quantify the financial impact of delays on order profitability.
- Identify primary operational bottlenecks driving late deliveries.
- Build a predictive model to flag high-risk orders before shipment.
- Deliver actionable recommendations to improve on-time delivery and profitability.

3. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline.

172,765 Total Orders	**94,523** Late Deliveries	**54.71%** Late Delivery %	**45.29%** On-Time Delivery %
$7.5M Total Profit	**$2.1M** Profit at Risk	**3 days** 90th Percentile Delay	**$22.03** Avg Order Profit

Late delivery is therefore a majority-experience rather than an edge-case event. The financial exposure is material, while the 90th percentile delay indicates that extreme lateness is generally contained within a few days.

4. Profitability Analysis

4.1 Profitability Distribution

Order-level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.

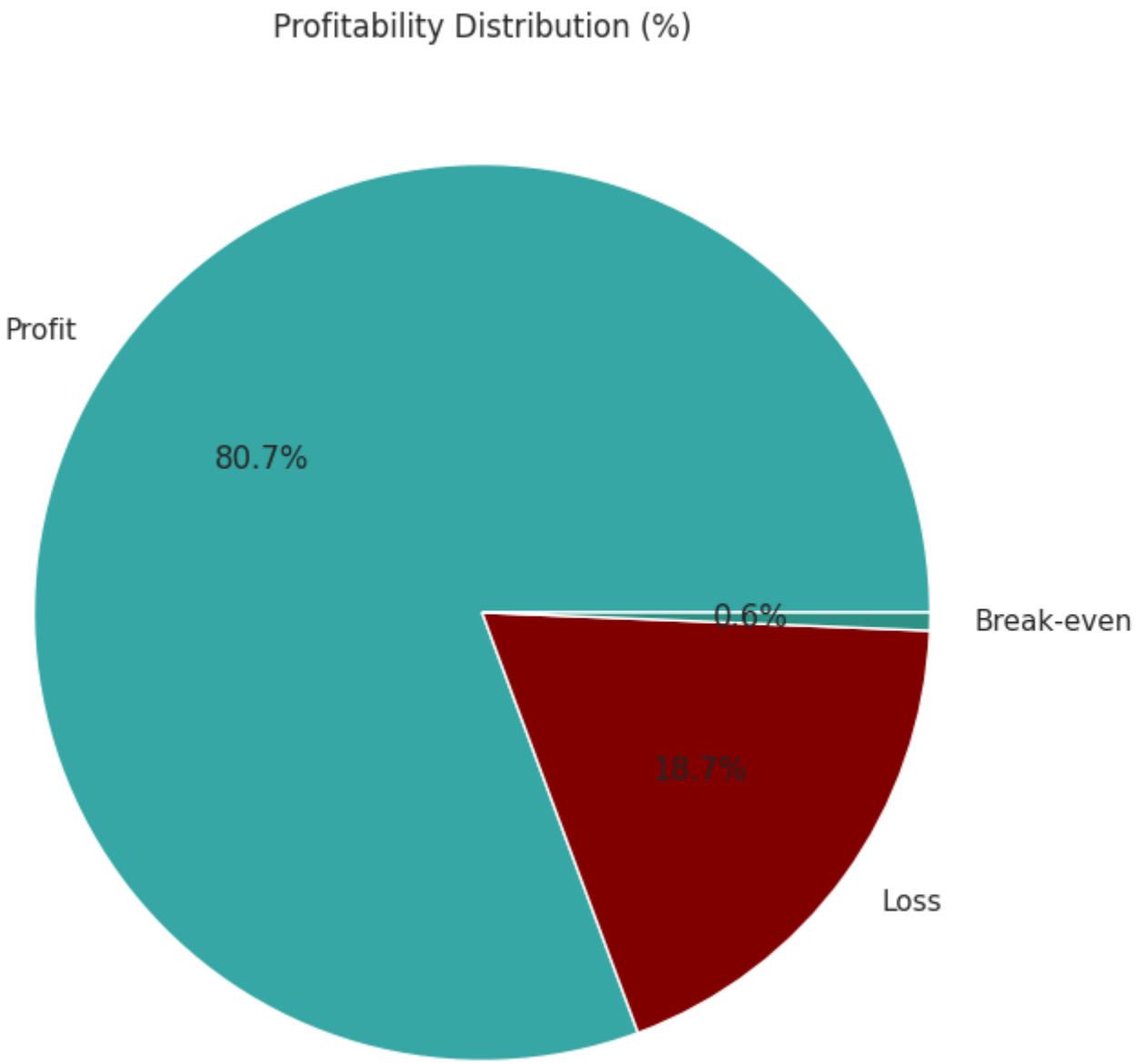


Figure 1: Profitability Distribution (%)

The distribution establishes that the dominant opportunity is not simply increasing sales volume; it is protecting profitable order economics from operational leakage caused by delays.

4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly 1 day late—the single largest cohort. Combined, orders delayed by 1–4 days account for 54.7% of all order volume, directly mapping to the overall late-delivery rate.

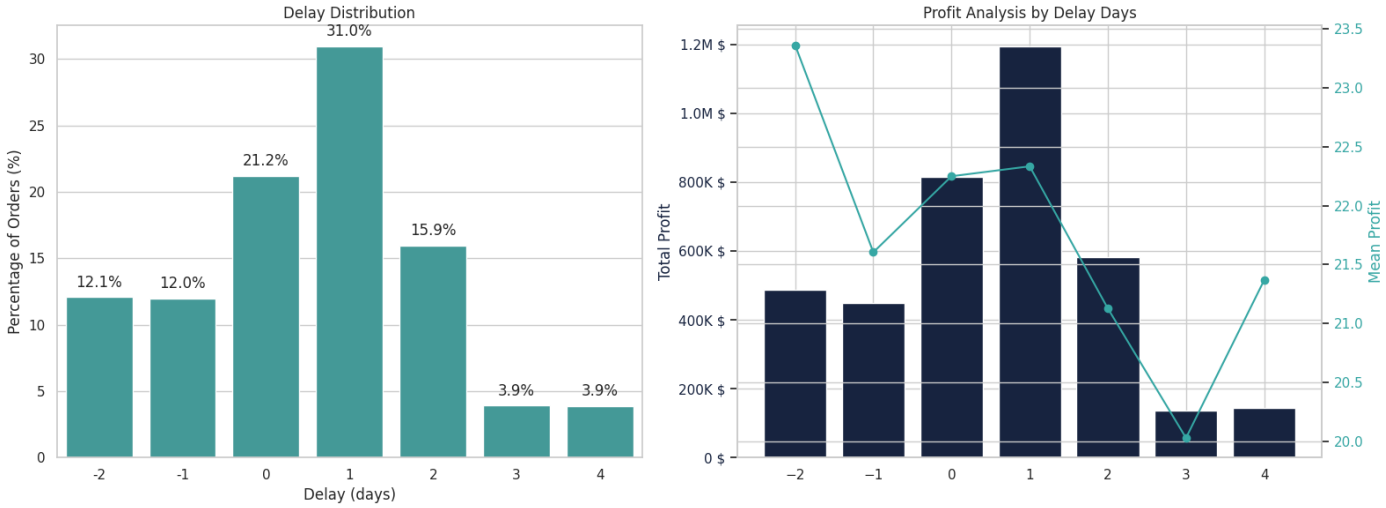


Figure 2: Delay Distribution (left) and Profit Analysis by Delay Days (right)

A critical observation: mean profit per order remains broadly stable at approximately $20–$23 across delay levels. The profitability problem is therefore driven by volume of delayed orders rather than a dramatic collapse in individual order economics, making systemic throughput improvement a high-leverage intervention.

5. Bottleneck Detection

Delay percentage was computed across six key operational dimensions. The charts reveal where the fulfillment process breaks down most severely.

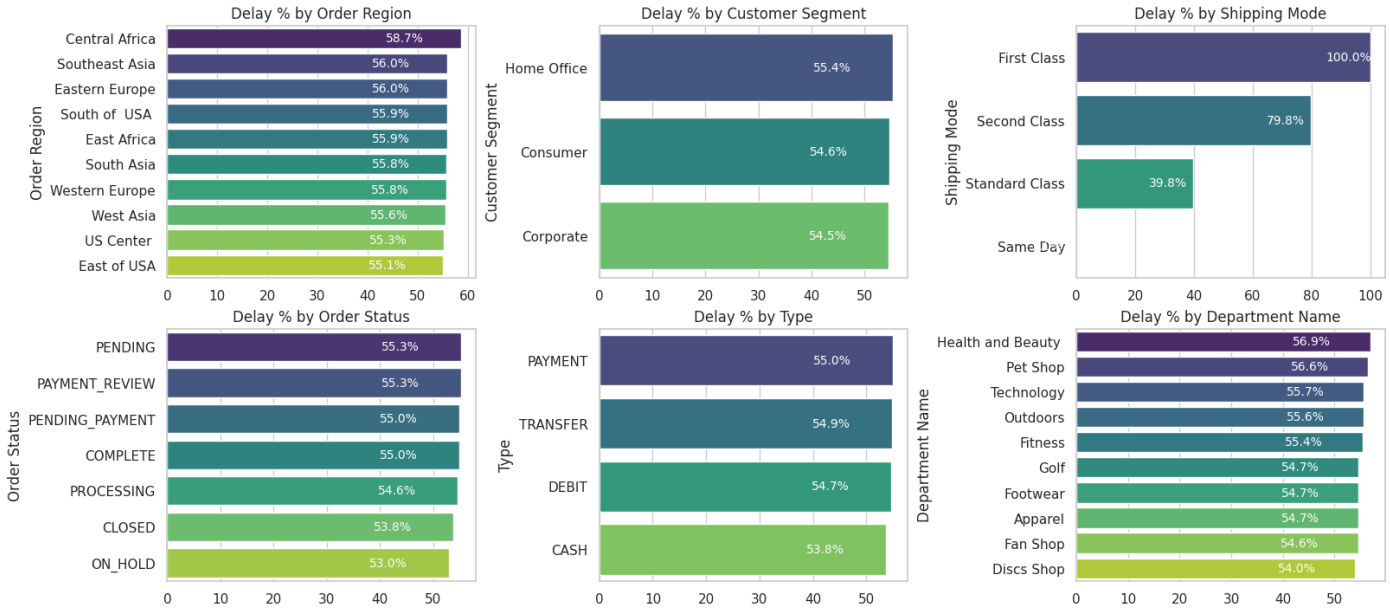


Figure 3: Delay % across six operational categories — Region, Segment, Shipping Mode, Order Status, Payment Type, Department

• **Shipping Mode is the #1 Lever**	First Class: 100% delay rate. Second Class: 79.8%. Standard Class: 39.8%. Same Day: 0%. Mode assignment should therefore be a primary operational audit area.
• **Regional Spread is Narrow (55–59%)**	Central Africa leads at 58.7%, but regions broadly cluster between 55–59%, indicating a systemic issue rather than a single-region-only problem.
• **Customer Segments Nearly Identical (54.5–55.4%)**	Consumer, Corporate and Home Office customers experience very similar delay rates, suggesting the issue is not concentrated in one customer segment.
• **Health & Beauty (56.9%) and Pet Shop (56.6%) Lead by Department**	These departments marginally exceed the broader department range and warrant targeted inventory, fulfillment and carrier review.

6. Root Cause Analysis

Central Africa was selected for deep-dive root cause analysis as the highest-delay region (58.7%). The chart ranks the top 10 driver factors by delay percentage within this region.

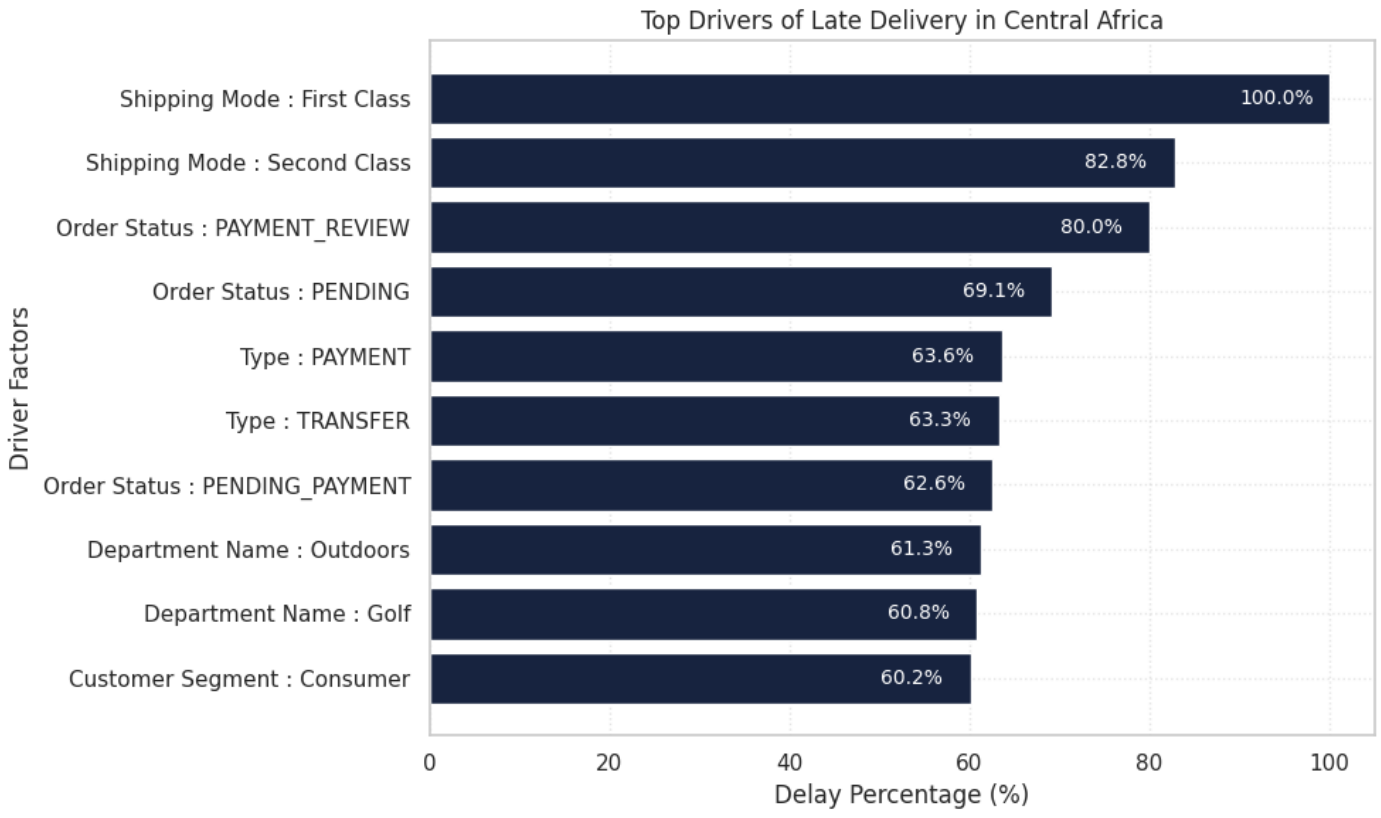


Figure 4: Top 10 Drivers of Late Delivery in Central Africa — ranked by delay percentage

The strongest observed drivers are shipping-mode outcomes, followed by payment-related order states and selected department/customer combinations. These factors identify concrete operational areas for intervention rather than merely describing the overall delay rate.

• **Highest observed driver**	First Class shipping in Central Africa shows a 100.0% delay rate, followed by Second Class at 82.8%.

7. Time-Based Delay Patterns

Three temporal dimensions were analysed: month of year, day of week, and hour of day. Delay rates are relatively stable across dimensions, but specific peaks highlight opportunities for targeted capacity planning.

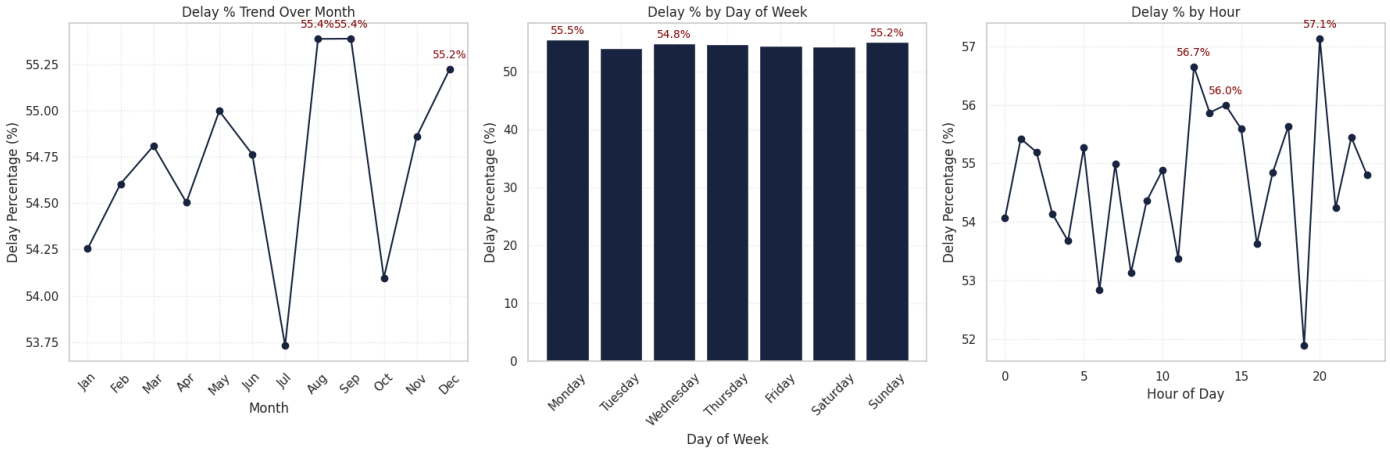


Figure 5: Delay % by Month (left), Day of Week (centre), and Hour of Day (right) — annotations mark top peaks

• Peak Delay Months: August & September (55.4%) and December (55.2%)	These months show the highest observed delay rates and can be used as planning triggers for seasonal capacity and carrier readiness.
• Day-of-Week Variance is Minimal	Monday (55.5%) and Sunday (55.2%) are among the highest, but the spread across the week is less than one percentage point.
• Intra-Day Peaks: Hour 21 (57.1%), Hour 11 (56.7%), Hour 12 (56.0%)	Late-evening and midday peaks suggest opportunities to review processing cutoffs, staffing and order-volume handling.

8. Machine Learning Model Results

A supervised classification model was built to predict whether an individual order will be delivered late (Late_delivery_risk = 1). The pipeline included frequency encoding of categorical variables, a stratified train/test split, SMOTE oversampling to address class imbalance, and Random Forest classification.

Class Imbalance Handling (SMOTE)

Split	Class 0 (On-time)	Class 1 (Late)
Before SMOTE (train)	59,030	79,182
After SMOTE (train)	79,182	79,182

Random Forest Classifier – Test Set Performance (34,553 records)

Metric	Class 0 (On-time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	**0.74**
Recall	0.72	0.75	**0.74**
F1-Score	0.70	0.77	**0.74**
Accuracy	—	—	**0.74**
Support	14,758	19,795	34,553

• **74% Overall Accuracy**	The model correctly classifies roughly three in four test orders.
• **Higher Precision on Late Orders (0.78)**	When the model predicts an order will be late, it is correct 78% of the time.
• **Recall for Late Orders at 0.75**	The model captures 75% of actual late orders.
• **Class 0 Harder to Predict (Precision 0.68)**	On-time deliveries share characteristics with late orders, reflecting the systemic nature of the delivery problem.

9. Strategic Recommendations

Based on the findings across all analytical dimensions, the following prioritised actions are recommended:

CRITICAL	**1. Immediately Audit First Class & Second Class Shipping Capacity** First Class shipping has a 100% delay rate and Second Class 79.8%. Conduct an immediate carrier SLA and mode-assignment audit.
HIGH	**2. Deploy the Predictive Alert System** The Random Forest model (74% accuracy; 78% precision on late orders) can support proactive risk identification and intervention.
HIGH	**3. Resolve Payment Processing Bottlenecks** PAYMENT_REVIEW and PENDING_PAYMENT states show elevated delay rates. Introduce automated escalation for stalled payment reviews.
MEDIUM	**4. Develop Seasonal Surge Capacity Plans** August, September and December are high-delay periods. Build pre-season capacity plans covering temporary logistics partnerships and inventory pre-positioning.
MEDIUM	**5. Default to Standard Class for Eligible Orders** Standard Class has a 39.8% delay rate versus materially higher rates for First and Second Class. Re-evaluate shipping-mode assignment logic.
MEDIUM	**6. Investigate High-Delay Departments in Africa** Outdoors (61.3%) and Golf (60.8%) in Central Africa show elevated delay rates. Audit stock availability, pick-and-pack time and carrier assignment.
LOW	**7. Reduce Loss-Making Order Share (18.7%)** Combine pricing/discount review with delay analysis to identify whether discounted orders are disproportionately late.
LOW	**8. Continuously Retrain the Predictive Model** Retrain quarterly and explore carrier-level, weather and warehouse-utilisation features to improve recall and robustness.

10. Conclusion

This analysis surfaces a clear operational picture: a global e-commerce operation where 54.71% of orders fail to meet their promised delivery windows, with approximately $2.1M in profit at risk. The scale of the delay problem makes delivery reliability a central business-performance issue.

The observed drivers are identifiable and actionable. Shipping-mode performance is the most pronounced operational signal, payment-processing states form a secondary bottleneck, and seasonal volume patterns provide a clear capacity-planning opportunity. Geographic differences exist, but the narrow regional spread indicates that the problem is broader than a single market.

A Random Forest predictive model trained on readily available order features achieves 74% accuracy in identifying at-risk orders. Used alongside operational controls, this creates a practical foundation for proactive late-delivery intervention.

Priority Area	Current State	Target State
Late Delivery Rate	54.71%	**< 30% within 12 months**
First Class On-Time Rate	0%	**> 80%**
Second Class On-Time Rate	20.2%	**> 60%**
Predictive Model Accuracy	74%	**> 82% (enhanced model)**
Loss-Making Orders	18.7%	**< 12%**
Profit at Risk	$2.1M	**Reduce by 40%**

With focused execution on the eight recommendations in this report—starting with the critical shipping-mode audit and predictive alert system deployment—the business can systematically improve delivery performance, protect its profit base, and build greater operational reliability for sustained e-commerce growth.